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Users module

Users knows who is calling. The PoC has no sign-in, so it has one built-in demo user.

Folder	`src/backend/app/users/`
Tables	`users`
Routes	None
Uses	Only the kernel
Used by	wallet and trading routes, the seed script
Issue	#279

Files

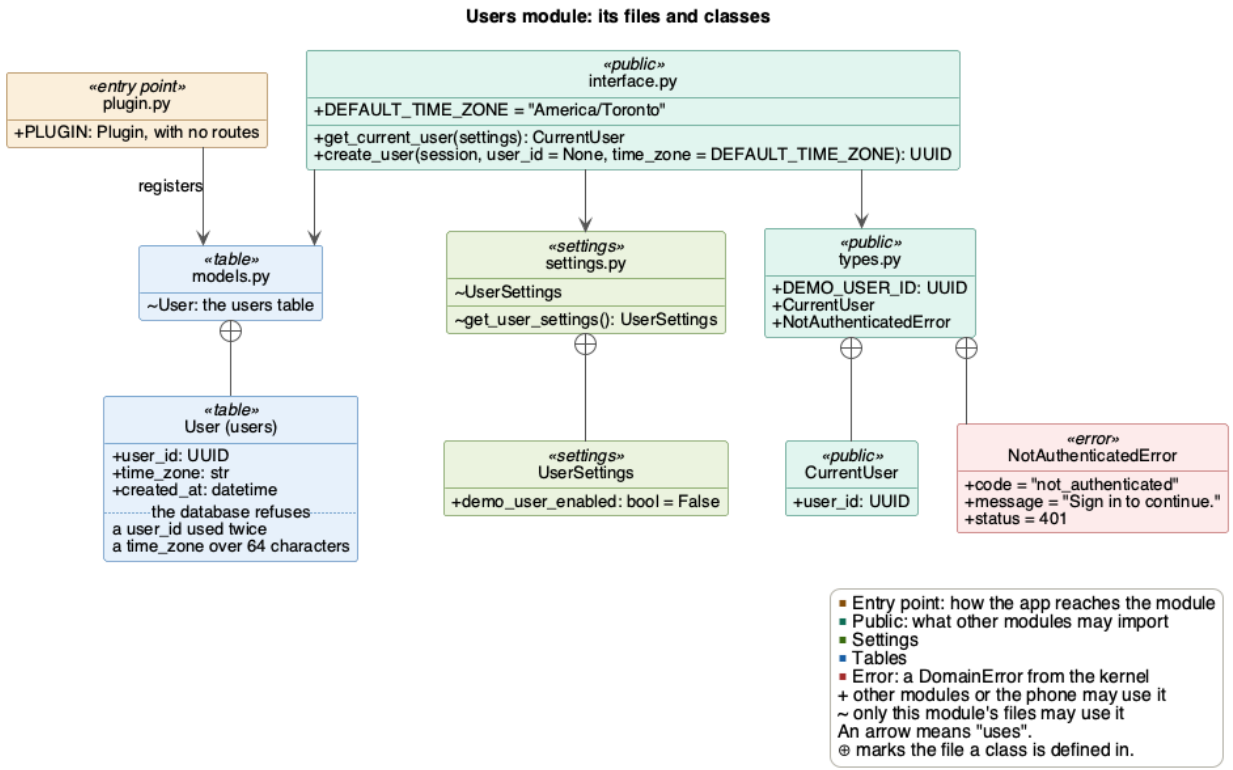


Figure 1: UML class diagram of the users module's files and classes

Other modules' routes use `get_current_user` and `CurrentUser`. The seed script uses `create_user` and `DEMO_USER_ID`. `plugin.py` has no routes yet. It only adds the `users` table to the app.

Who is calling

`get_current_user` is the one place that knows who is calling. A route that needs a user asks for one with `Depends(get_current_user)`, so the phone never sends a user ID.

The PoC has no sign-in: - While `DEMO_USER_ENABLED` is on, every request is the demo user, whose ID is `DEMO_USER_ID`. The ID is fixed, so the seed script and `get_current_user` always agree on who it is. - While it is off, as it is by default, a route that needs a user answers 401 `not_authenticated` ("Sign in to continue."). A server without the setting never runs as the demo user by accident.

`DEMO_USER_ENABLED` is read from the environment or `.env`, once per process. `.env.example` turns it on, and `src/backend/README.md` has developers copy it to `.env`, so a development server runs as the demo user.

Real sign-in later only changes the inside of `get_current_user`. The routes keep asking for the user the same way.

The demo user must exist first

`get_current_user` doesn't check that the user exists. The demo user and its live portfolio have to be made first, or wallet's functions fail for it. The seed script does both, with `create_user(session, DEMO_USER_ID)` and `create_live_portfolio(session, DEMO_USER_ID)`. Run it from `src/backend`:

```
uv run python -m app.scripts.seed_demo
uv run python -m app.scripts.seed_demo reset
```

- Without `reset`, it adds whichever of the two is missing. Running it again changes nothing.
- With `reset`, it gives the demo user a new live portfolio at 0. The old one is deleted with everything under it, such as its balance changes and trades.

It uses only users' and wallet's public files, never their tables.

Adding a user

`create_user(session, user_id = None, time_zone = DEFAULT_TIME_ZONE)` adds a user and returns their ID. It flushes but never commits. - Without a `user_id`, the database makes one. The flush is what lets it return that ID. - A `user_id` that is already used fails at once, with `IntegrityError`. So the seed script adds the demo user in a savepoint, which undoes only that insert when the demo user already exists, such as after a failed run. - It refuses a time zone that doesn't exist with `ValueError`, before adding anything, so a stored zone always works. The `users` table doesn't check it. - Nothing reads the stored zone yet. Activity (#190) counts each day in the zone the phone sends with each sync, since a day follows where the user is (SEC-23). - The default, `DEFAULT_TIME_ZONE`, is `America/Toronto`, since the team and its first users are in Ontario.

Design patterns

- **Plug-in:** `plugin.py` adds users to the app, with no routes. The kernel doesn't change.
- **Dependency injection:** routes get the user through `Depends(get_current_user)`, and it gets its settings through `Depends(get_user_settings)`. Tests replace either in `app.dependency_overrides`, to pick the user or turn the demo user on.
- **Read-only records:** other modules get a frozen `CurrentUser`, never the `users` row.

Not built in the PoC

- Sign-in and accounts (FR_AcctCreate, FR_AcctSignIn). They bring users' routes, and the sign-in fields of the `users` table, such as the email address.

Tests

File	What it checks
`test_users.py`	The demo user when it is on, that it is off by default, `create_user` with its time zone check, and the table refusing a user with no time zone
`test_seed_demo.py`	The seed script: the demo user with a portfolio at 0, running it twice, a user with no portfolio, and `reset`